IMPLICATIONS BETWEEN EXACTNESS PROPERTIES OF A MORPHISM OF TOPOI

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument for the eight groups of implications; the sentence 'This is immediate from the definitions' concerns the first group, whose content is a definitional consequence, and the last group is attributed to a named categorical result, both preserved verbatim inside the counted span.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- Characterization of monomorphisms and epimorphisms by the diagonal and the codiagonal. Categories, Lemma categories-lemma-characterize-mono-epi. Rewrites injectivity and surjectivity as isomorphisms statements used throughout the proof.
- Full faithfulness of an adjoint pair. Categories, Lemma categories-lemma-adjoint-fully-faithful. Supplies the equivalence between the counit and unit being an isomorphism and full faithfulness.
- The pullback of a morphism of topoi is exact. Same chapter, the definition of a morphism of topoi. Allows the pullback to be applied to the diagrams considered in the proof.

Source reference. Original lemma 08LR: sources/stacks-sites/pinned-original-source.tex, source lines 9867–9867. Navigation pointer invoked at proof line 9867; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 086K: sources/stacks-sites/pinned-original-source.tex, source lines 9883–9883. Navigation pointer invoked at proof line 9883; the counted span contains the pointer only and the cited result is not reproduced inside it, so

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Source reference. Original lemma 07RB: sources/stacks-sites/pinned-original-source.tex, source lines 9959–9959. Navigation pointer invoked at proof line 9959; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

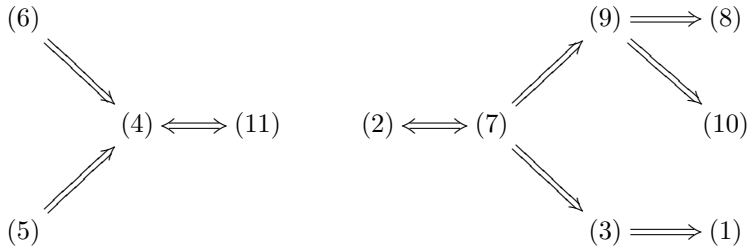
Lemma 0.1. *Let $f : Sh(\mathcal{C}) \to Sh(\mathcal{D})$ be a morphism of topoi. Consider the following properties (on sheaves of sets):*

(1) *f_* is faithful,*
(2) *f_* is fully faithful,*
(3) *$f^{-1}f_*\mathcal{F} \to \mathcal{F}$ is surjective for all $\mathcal{F}$ in $Sh(\mathcal{C})$,*
(4) *f_* transforms surjections into surjections,*
(5) *f_* commutes with coequalizers,*
(6) *f_* commutes with pushouts,*
(7) *$f^{-1}f_*\mathcal{F} \to \mathcal{F}$ is an isomorphism for all $\mathcal{F}$ in $Sh(\mathcal{C})$,*
(8) *f_* reflects injections,*
(9) *f_* reflects surjections,*
(10) *f_* reflects bijections, and*
(11) *for any surjection $\mathcal{F} \to f^{-1}\mathcal{G}$ there exists a surjection $\mathcal{G}' \to \mathcal{G}$ such that $f^{-1}\mathcal{G}' \to f^{-1}\mathcal{G}$ factors through $\mathcal{F} \to f^{-1}\mathcal{G}$.*

Then we have the following implications

(a) *(2) $\Rightarrow$ (1),*
(b) *(3) $\Rightarrow$ (1),*
(c) *(7) $\Rightarrow$ (1), (2), (3), (8), (9), (10).*
(d) *(3) $\Leftrightarrow$ (9),*
(e) *(6) $\Rightarrow$ (4) and (5) $\Rightarrow$ (4),*
(f) *(4) $\Leftrightarrow$ (11),*
(g) *(9) $\Rightarrow$ (8), (10), and*
(h) *(2) $\Leftrightarrow$ (7).*

Picture



Proof. Proof of (a): This is immediate from the definitions.

Proof of (b). Suppose that $a, b : \mathcal{F} \to \mathcal{F}'$ are maps of sheaves on $\mathcal{C}$. If $f_*a = f_*b$, then $f^{-1}f_*a = f^{-1}f_*b$. Consider the commutative diagram

$$\begin{array}{ccc} \mathcal{F} & \rightrightarrows & \mathcal{F}' \\ \uparrow & & \uparrow \\ f^{-1}f_*\mathcal{F} & \rightrightarrows & f^{-1}f_*\mathcal{F}' \end{array}$$

If the bottom two arrows are equal and the vertical arrows are surjective then the top two arrows are equal. Hence (b) follows.

Proof of (c). Suppose that $a : \mathcal{F} \to \mathcal{F}'$ is a map of sheaves on $\mathcal{C}$. Consider the commutative diagram

$$\begin{array}{ccc} \mathcal{F} & \longrightarrow & \mathcal{F}' \\ \uparrow & & \uparrow \\ f^{-1}f_*\mathcal{F} & \longrightarrow & f^{-1}f_*\mathcal{F}' \end{array}$$

If (7) holds, then the vertical arrows are isomorphisms. Hence if f_*a is injective (resp. surjective, resp. bijective) then the bottom arrow is injective (resp. surjective, resp. bijective) and hence the top arrow is injective (resp. surjective, resp. bijective). Thus we see that (7) implies (8), (9), (10). It is clear that (7) implies (3). The implications (7) $\Rightarrow$ (2), (1) follow from (a) and (h) which we will see below.

Proof of (d). Assume (3). Suppose that $a : \mathcal{F} \to \mathcal{F}'$ is a map of sheaves on $\mathcal{C}$ such that f_*a is surjective. As f^{-1} is exact this implies that $f^{-1}f_*a : f^{-1}f_*\mathcal{F} \to f^{-1}f_*\mathcal{F}'$ is surjective. Combined with (3) this implies that a is surjective. This means that (9) holds. Assume (9). Let $\mathcal{F}$ be a sheaf on $\mathcal{C}$. We have to show that the map $f^{-1}f_*\mathcal{F} \to \mathcal{F}$ is surjective. It suffices to show that $f_*f^{-1}f_*\mathcal{F} \to f_*\mathcal{F}$ is surjective. And this is true because there is a canonical map $f_*\mathcal{F} \to f_*f^{-1}f_*\mathcal{F}$ which is a one-sided inverse.

Proof of (e). We use Categories, Lemma 08LR without further mention. If $\mathcal{F} \to \mathcal{F}'$ is surjective then $\mathcal{F}' \amalg_{\mathcal{F}} \mathcal{F}' \to \mathcal{F}'$ is an isomorphism. Hence (6) implies that

$$f_*\mathcal{F}' \amalg_{f_*\mathcal{F}} f_*\mathcal{F}' = f_*(\mathcal{F}' \amalg_{\mathcal{F}} \mathcal{F}') \longrightarrow f_*\mathcal{F}'$$

is an isomorphism also. And this in turn implies that $f_*\mathcal{F} \to f_*\mathcal{F}'$ is surjective. Hence we see that (6) implies (4). If $\mathcal{F} \to \mathcal{F}'$ is surjective then $\mathcal{F}'$ is the coequalizer of the two projections $\mathcal{F} \times_{\mathcal{F}'} \mathcal{F} \to \mathcal{F}$ by Lemma 086K. Hence if (5) holds, then $f_*\mathcal{F}'$ is the coequalizer of the two projections

$$f_*(\mathcal{F} \times_{\mathcal{F}'} \mathcal{F}) = f_*\mathcal{F} \times_{f_*\mathcal{F}'} f_*\mathcal{F} \longrightarrow f_*\mathcal{F}$$

which clearly means that $f_*\mathcal{F} \to f_*\mathcal{F}'$ is surjective. Hence (5) implies (4) as well.

Proof of (f). Assume (4). Let $\mathcal{F} \to f^{-1}\mathcal{G}$ be a surjective map of sheaves on $\mathcal{C}$. By (4) we see that $f_*\mathcal{F} \to f_*f^{-1}\mathcal{G}$ is surjective. Let $\mathcal{G}'$ be the fibre product

$$\begin{array}{ccc} f_*\mathcal{F} & \longrightarrow & f_*f^{-1}\mathcal{G} \\ \uparrow & & \uparrow \\ \mathcal{G}' & \longrightarrow & \mathcal{G} \end{array}$$

so that $\mathcal{G}' \to \mathcal{G}$ is surjective also. Consider the commutative diagram

$$\begin{array}{ccc} \mathcal{F} & \longrightarrow & f^{-1}\mathcal{G} \\ \uparrow & & \uparrow \\ f^{-1}f_*\mathcal{F} & \longrightarrow & f^{-1}f_*f^{-1}\mathcal{G} \\ \uparrow & & \uparrow \\ f^{-1}\mathcal{G}' & \longrightarrow & f^{-1}\mathcal{G} \end{array}$$

and we see the required result. Conversely, assume (11). Let $a : \mathcal{F} \to \mathcal{F}'$ be surjective map of sheaves on $\mathcal{C}$. Consider the fibre product diagram

$$\begin{array}{ccc} \mathcal{F} & \longrightarrow & \mathcal{F}' \\ \uparrow & & \uparrow \\ \mathcal{F}'' & \longrightarrow & f^{-1}f_*\mathcal{F}' \end{array}$$

Because the lower horizontal arrow is surjective and by (11) we can find a surjection $\gamma : \mathcal{G}' \to f_*\mathcal{F}'$ such that $f^{-1}\gamma$ factors through $\mathcal{F}'' \to f^{-1}f_*\mathcal{F}'$:

$$\begin{array}{ccccc} & & \mathcal{F} & \longrightarrow & \mathcal{F}' \\ & & \uparrow & & \uparrow \\ f^{-1}\mathcal{G}' & \longrightarrow & \mathcal{F}'' & \longrightarrow & f^{-1}f_*\mathcal{F}' \end{array}$$

Pushing this down using f_* we get a commutative diagram

$$\begin{array}{ccccc} & & f_*\mathcal{F} & \longrightarrow & f_*\mathcal{F}' \\ & & \uparrow & & \uparrow \\ f_*f^{-1}\mathcal{G}' & \longrightarrow & f_*\mathcal{F}'' & \longrightarrow & f_*f^{-1}f_*\mathcal{F}' \\ \uparrow & & & & \uparrow \\ \mathcal{G}' & & \longrightarrow & & f_*\mathcal{F}' \end{array}$$

which proves that (4) holds.

Proof of (g). Assume (9). We use Categories, Lemma 08LR without further mention. Let $a : \mathcal{F} \to \mathcal{F}'$ be a map of sheaves on $\mathcal{C}$ such that f_*a is injective. This means that $f_*\mathcal{F} \to f_*\mathcal{F} \times_{f_*\mathcal{F}'} f_*\mathcal{F} = f_*(\mathcal{F} \times_{\mathcal{F}'} \mathcal{F})$ is an isomorphism. Thus by (9) we see that $\mathcal{F} \to \mathcal{F} \times_{\mathcal{F}'} \mathcal{F}$ is surjective, i.e., an isomorphism. Thus a is injective, i.e., (8) holds. Since (10) is trivially equivalent to (8) + (9) we are done with (g).

Proof of (h). This is Categories, Lemma 07RB. □